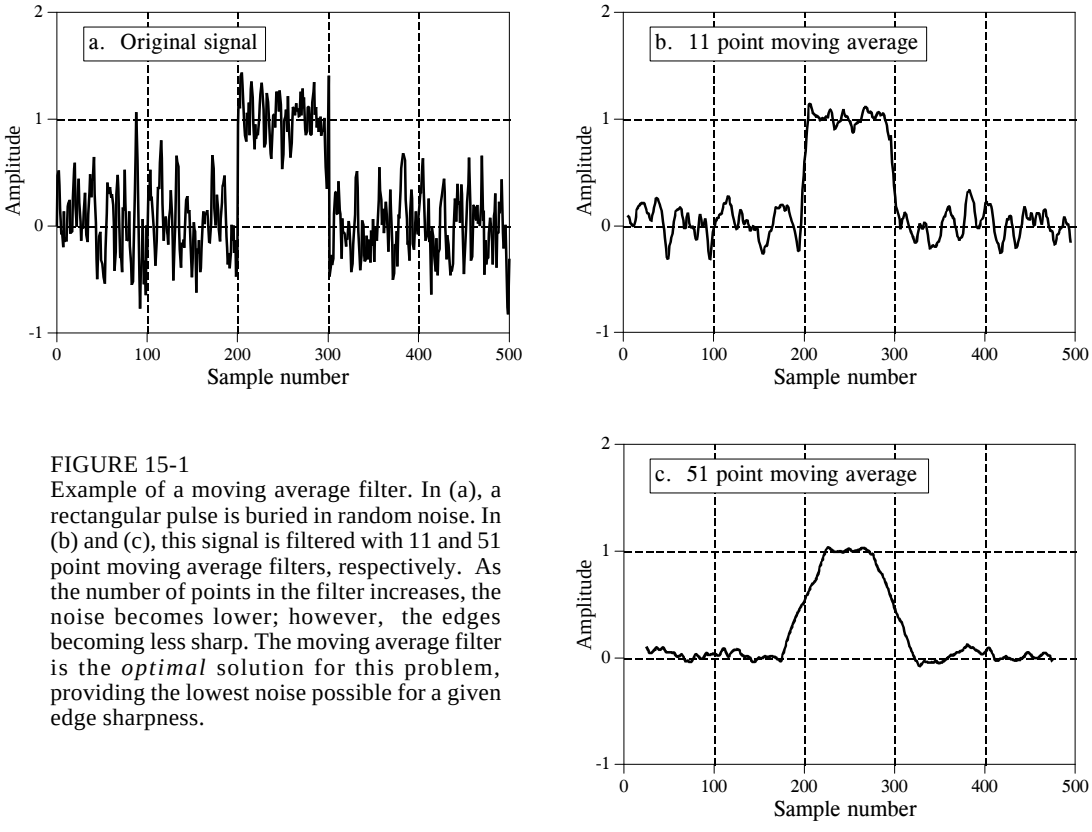


FIGURE 15-1
Example of a moving average filter. In (a), a rectangular pulse is buried in random noise. In (b) and (c), this signal is filtered with 11 and 51 point moving average filters, respectively. As the number of points in the filter increases, the noise becomes lower; however, the edges becoming less sharp. The moving average filter is the *optimal* solution for this problem, providing the lowest noise possible for a given edge sharpness.

Figure 15-1 shows an example of how this works. The signal in (a) is a pulse buried in random noise. In (b) and (c), the smoothing action of the moving average filter decreases the amplitude of the random noise (good), but also reduces the sharpness of the edges (bad). Of all the possible linear filters that could be used, the moving average produces the lowest noise for a given edge sharpness. The amount of noise reduction is equal to the square-root of the number of points in the average. For example, a 100 point moving average filter reduces the noise by a factor of 10.

To understand why the moving average if the best solution, imagine we want to design a filter with a fixed edge sharpness. For example, let's assume we fix the edge sharpness by specifying that there are eleven points in the rise of the step response. This requires that the filter kernel have eleven points. The optimization question is: how do we choose the eleven values in the filter kernel to minimize the noise on the output signal? Since the noise we are trying to reduce is random, none of the input points is special; each is just as noisy as its neighbor. Therefore, it is useless to give preferential treatment to any one of the input points by assigning it a larger coefficient in the filter kernel. The lowest noise is obtained when all the input samples are treated equally, i.e., the moving average filter. (Later in this chapter we show that other filters are essentially *as* good. The point is, no filter is *better* than the simple moving average).